

	Analytic	Electric Network	Random Walk	Linear Algebra
Laplacian $\Delta f(x)$	$\left(\sum_{y \sim x} \frac{c_{xy}}{c_x} f(y) \right) - f(x)$ (difference between weighted average of f at neighbours of x and value of f at x)	$-\frac{i_x}{c_x}$ ($f(z)$ is voltage at z , $i_x = \sum_y c_{xy}(f(x) - f(y))$ is current out of x)	$\mathbb{E}^x f(X_1) - f(x)$	$\Delta = D^{-1}A = -(I - P)$ (P is transition matrix, D is diagonal with $D_{xx} = c_x$, A is conductance matrix)
Dirichlet Problem G is “boundary”, $H = V \setminus G$ is “interior”	$f = g$ on G , $\Delta f = 0$ on H (uniqueness and hence existence from max pple)	f gives voltages on H for applied voltages g on G	$f(x) = \mathbb{E}^x g(X_{T_G})$ ($= P^x(T_a < T_b)$ if $G = \{a, b\}$)	$f_H = -A_{HH}^{-1}A_{HG}g$ where $A = \begin{bmatrix} A_{GG} & A_{GH} \\ A_{HG} & A_{HH} \end{bmatrix}$ (solve $f_G = g$, $D^{-1}A_{HG}f_G + D^{-1}A_{HH}f_H = 0$)
Green’s Function $\mathcal{G}_a^G$ (for fixed G and $a \in H = G^c$)	$-\Delta \mathcal{G}_a^G = \frac{1}{c_a} \chi_a \text{ on } H$ $\mathcal{G}_a^G = 0 \text{ on } G$ (uniqueness and hence existence from max pple)	$\mathcal{G}_a^G$ is harmonic extension of voltage at a and zero voltage at G s.t. <i>unit current</i> flows from a to G	$\mathcal{G}_a^G(x) = \frac{1}{c_x} \mathbb{E}^a V_{x \rightarrow G}$ $= \frac{1}{c_{xz}} \mathbb{E}^a \text{Tr}_{xz \rightarrow G} \text{ if } z \sim x$ $= \mathbb{R}_{aG} P^x(T_a < T_G)$ (verify eqns for first and third except at a , evaluate both at a ; second follows from first with $\text{Tr}_{xz \rightarrow G}$ the no of traverses of xz on way to G)	$\mathcal{G}_a^G _H = -[A_{HH}^{-1}]_a,$ where $[M]_a$ is col a of matrix M (solve $(I_H - P_{HH})\mathcal{G}_a^G = [D_H^{-1}]_a$)
Effective Resistance $\mathbb{R}_{aG}$ ($G = \{b\}$ gives a <i>metric</i> $\mathbb{R}_{ab}$)	$\mathbb{R}_{aG} = \mathcal{G}_a^G(a)$ (Green’s function for bdy G)	<i>Defn:</i> $\mathbb{R}_{aG} := \mathcal{G}_a^G(a)$ is voltage at a s.t. zero voltage at G induces a unit flow from a to G	$\mathbb{R}_{aG} = \frac{1}{c_a} \mathbb{E}^a V_{a \rightarrow G}$ $= (c_a P^a(T_G < T_a^+))^{-1}$ (the first equality by defn, the second by standard prob thy)	$\mathbb{R}_{aG} = -(A_{HH}^{-1})_{aa}$